NumPy Tutorial 1: Introduction and Basic Concepts

This chapter introduces NumPy in a clear and beginner friendly way. After completing this tutorial, students will understand what NumPy is, why it is used in Data Science, and how to start working with it.

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1. **Introduction to NumPy**

NumPy stands for **Numerical Python**. It is a powerful Python library used for numerical computing and working with arrays.

NumPy is mainly used in Data Science, Machine Learning, Scientific Computing, and Data Analysis.

NumPy allows us to work with large amounts of numerical data in a very fast and efficient way.

NumPy works very well with other libraries such as Pandas, Matplotlib, and Scikit Learn.

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2. **Why We Use NumPy in Data Science**

In real world projects, we deal with large amounts of numerical data. Working with this data using normal Python lists is very slow and difficult.

NumPy makes this process simple, fast, and efficient. It provides built in functions to:

- Create and store large arrays of numbers.
- Perform mathematical operations on entire arrays at once.
- Reshape and manipulate data easily.
- Perform statistical calculations quickly.
- Work with multi-dimensional data.

Because of these features, NumPy is the foundation of almost every Data Science library.

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3. **NumPy vs Python Lists**

You might ask — why use NumPy when Python already has lists?

Here is the difference:

Feature	Python List	NumPy Array
Speed	Slow	Very Fast
Memory	Uses more memory	Uses less memory
Math operations	Manual loops needed	Direct operations

Feature	Python List	NumPy Array
Data type	Mixed types allowed	Same type only
Size	Can grow dynamically	Fixed size

NumPy is **50 times faster** than Python lists for numerical operations. This is why every Data Scientist uses NumPy.

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4. **Installing NumPy**

If you are using a Conda environment, activate your environment first:

```
conda activate eda_env
```

Then install NumPy:

```
conda install numpy
```

Or using pip:

```
pip install numpy
```

After installation, you can import it in Python.

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5. **Importing NumPy**

In Python, we usually import NumPy using the alias np.

```
import numpy as np
```

The name np is a standard convention used in almost all Data Science projects.

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6. **What is a NumPy Array**

The most important data structure in NumPy is the **ndarray** — which stands for N-dimensional array.

An array is simply a collection of numbers stored in an organized way.

NumPy arrays can be:

- **1D array** — a single row of numbers (like a list)
- **2D array** — rows and columns (like a table)
- **3D array** — multiple tables (like a cube of data)

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7. **Creating Your First NumPy Array**

We can create a NumPy array from a Python list.

```
import numpy as np

numbers = [10, 20, 30, 40, 50]
arr = np.array(numbers)

print(arr)
```

Output:

```
[10 20 30 40 50]
```

In this example:

- numbers is a normal Python list.
- np.array converts it into a NumPy array.
- Notice there are no commas in the output — this is how NumPy displays arrays.

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8. **Checking Array Information**

After creating an array, we often check its basic information.

Data Type

```
print(arr.dtype)
```

This shows what type of data is stored — int, float, etc.

Shape

```
print(arr.shape)
```

This shows the number of rows and columns.

Size

```
print(arr.size)
```

This shows the total number of elements.

Number of Dimensions

```
print(arr.ndim)
```

This shows how many dimensions the array has.

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9. **NumPy Array Data Types**

NumPy stores data in specific types. The most common ones are:

Data Type	Description	Example
int32 / int64	Integer numbers	1, 5, 100
float32 / float64	Decimal numbers	3.14, 2.5
bool	True or False	True, False
str	Text	"hello"

NumPy automatically selects the best data type. But we can also specify it manually.

```
arr = np.array([1, 2, 3], dtype=float)
print(arr)
```

Output:

```
[1. 2. 3.]
```

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10. **Why NumPy is the Foundation of Data Science**

Almost every Data Science library is built on top of NumPy.

- **Pandas** uses NumPy arrays internally for storing data.
- **Matplotlib** uses NumPy arrays for plotting graphs.
- **Scikit Learn** uses NumPy arrays for Machine Learning models.
- **TensorFlow and PyTorch** use NumPy style arrays for Deep Learning.

This is why learning NumPy properly is very important before moving to advanced topics.

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11. **Conclusion of Tutorial 1**

In this tutorial, students have learned:

1. What NumPy is and why it is important.
2. How NumPy is faster and better than Python lists.
3. How to install and import NumPy.
4. What a NumPy array is and its types.
5. How to create a basic array and check its information.
6. Why NumPy is the foundation of Data Science.

In the next tutorial, we will learn how to create different types of arrays — 1D, 2D, and 3D — and explore special array creation functions.

This completes NumPy Tutorial 1.